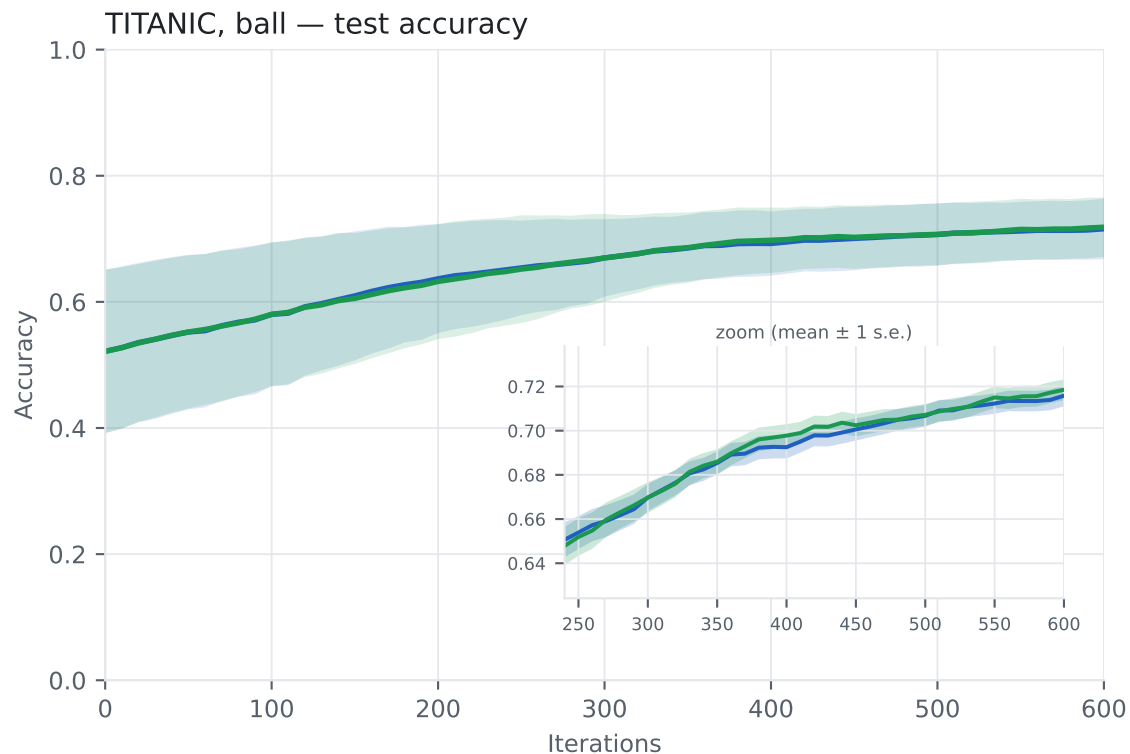
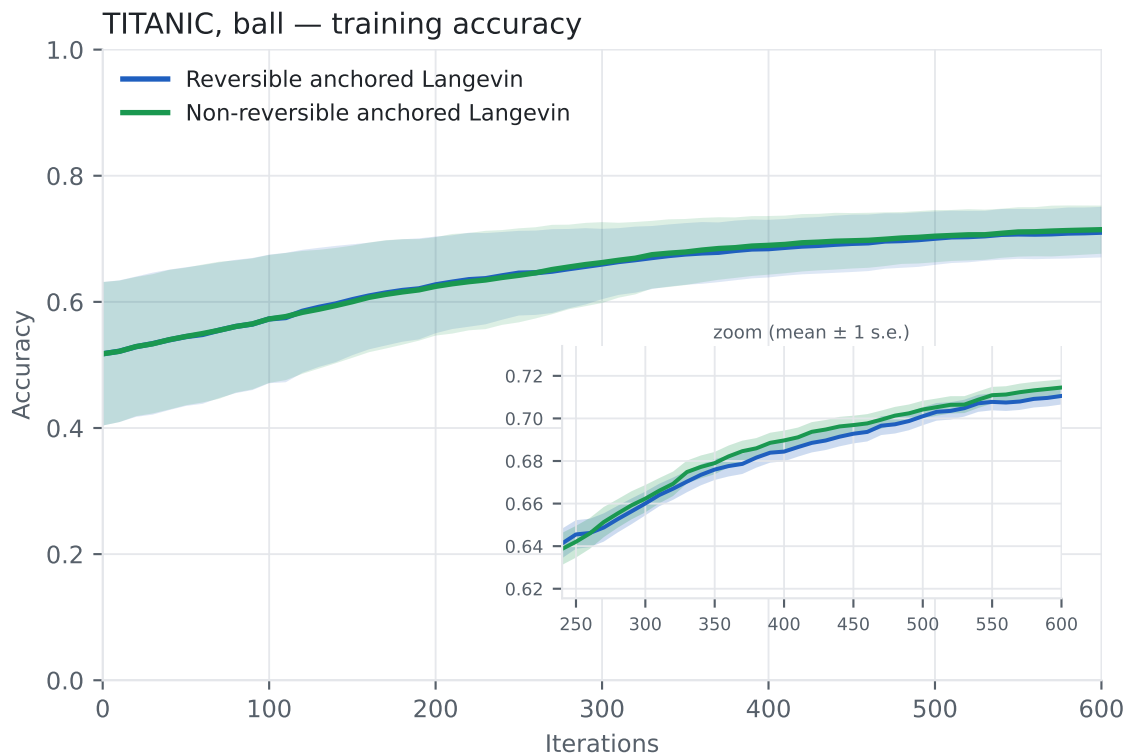


titanic_ball: exact gradient, LASSO anchor (first 600 iterations)



ALL RUNS STOPPED AT ITERATION 600 -- a truncation of the notebook's longer runs (identical W_0 and Gaussian stream), not a separate experiment. TITANIC -- ball $\|w\|^2 \leq 2$ (radius $\sqrt{2}$). $d = 10$ (intercept w_0 + nine features), no mini-batching: the EXACT gradient is used at every step. $\sigma = 10$, $\lambda = 7.12$, $\delta = 0.01082$, so a in $[0.50, 1]$. Block strengths $s = (5, 5, 5)$ on $w_{1..w_9}$ only. Step $\eta = 1e-05$. $R = 100$ coupled replicates (same W_0 , same Gaussian stream) on one fixed stratified 80/20 split ($n_{\text{train}} = 712$, $n_{\text{test}} = 179$). Paired difference AT ITERATION 600, non-reversible minus reversible: training $+0.0039$ ($t = +0.93$), test $+0.0027$ ($t = +0.42$). Bands: mean $\pm$ 1 SD ($\text{ddof} = 1$) across replicates -- repeat-run variability at a fixed split, not a confidence or credible interval.